

FACTSHEET FS19 · Robustness

Is it a bug, a few lucky episodes, or a sector bet?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · 2013–2026, momentum crashes also 1999–2013

Key facts	
Pre-registration	planning/PREREG_FS19.md, saved to the Factsheets folder before any FS19 number was computed; no deviations
A. Engine tests	truncation (look-ahead), iid-noise, cheat-timing, cost accounting, beta neutrality
B. Stress episodes	2015–16 sell-off, Q4 2018, COVID crash, COVID rebound, 2022 rate shock; momentum-crash regression (Daniel & Moskowitz 2016)
C. Sector neutrality	low beta and 12-1 momentum ranked within 11 sectors (static Morningstar/Yahoo map); primary test of FS19
Primary test	pooled sector-neutral shortlist (US, EU, UK, Denmark, SCANDI): NW $t > 2$ and at least half the unrestricted return

In one paragraph. The engine is clean and the result is not a sector bet, but "beta-neutral" low beta is not market-neutral. The signals are identical when rebuilt from truncated data, they earn nothing on random noise, and the cost arithmetic adds up to the last decimal. Ranked within sectors, the shortlist keeps **+5.3% a year (t 2.96)** of its unrestricted +8.0%, so the primary test is **passed**. The warning is in the crash tests. The low-beta book, scaled to beta-neutral with estimated betas, still carries a realised market beta of about +0.3 to +0.6 in the US, UK, EU and World, because low betas drift up and high betas drift down. It lost 29% in the US in the COVID crash and 24% in 2022. Momentum did the hedging. The FS18 overlay as a whole stayed within $\pm 8\%$ in every episode.

A. Engine tests

Test	What	Result	Verdict
1. Truncation (look-ahead)	19 signals rebuilt from data cut at Dec 2015, 2019, 2023 (US, EU, UK): 171 comparisons	identical signal values and name counts	pass
2. Noise	19 signals on iid noise, 5 seeds $\times$ US, EU: 190 gross L/S t-statistics	4.2% have $ t > 1.96$ (expected 5%); max $ t $ 2.65	pass
3. Cheat (timing)	next month's return as the signal vs this month's return	next month: $t \geq 189$; this month: $ t \leq 1.9$	pass
4. Cost accounting	gross – net = cost + borrow + financing; cost = 10 bp $\times$ 2 $\times$ turnover	largest error 3e-17	pass
5. Beta neutrality	realised beta of the 'beta-neutral' books on the equal-weight universe	flagged if $ \text{beta} > 0.3$: US low beta (+0.48), EU 12-1 momentum (-0.33), UK low beta (+0.35), World low beta (+0.58)	flag

Market	Low beta (beta-neutral): realised beta	12-1 momentum (beta-neutral): realised beta
US	+0.48	-0.14
EU	+0.28	-0.33
UK	+0.35	-0.23
Denmark	+0.02	-0.17
SCANDI	+0.07	-0.17
World	+0.58	-0.22

Realised beta: full-sample regression of the gross book return on the equal-weight universe, 2013–2026. The legs are scaled with 12-month ex-ante betas, clipped to 0.2–3. Frazzini & Pedersen (2014) shrink betas towards 1 for exactly this reason; our books do not, so "beta-neutral" means ex ante, not realised.

B. Stress episodes

Cumulative return in each episode, % (net of costs at \$50m per book)						
US	low beta	+27	+4	-29	-12	-24
	12-1 momentum	+9	-4	+9	+2	+10
US	overlay	+8	+2	+2	-4	+4
	low beta	+8	+10	-16	-6	-28
EU	12-1 momentum	+29	-8	+8	-3	-6
	overlay	+8	+1	-1	-1	-6
UK	low beta	+12	-13	-12	-2	-17
	12-1 momentum	+5	-5	+2	-3	+1
UK	overlay	+5	-5	-2	+1	-2
Denmark	low beta	+0	-16	-5	-21	+12
	12-1 momentum	+19	-4	+8	+1	-17
Denmark	overlay	+5	-6	-0	-5	-1
SCANDI	low beta	+4	-2	-11	-8	+1
	12-1 momentum	+15	-4	+5	-2	-9
SCANDI	overlay	+8	-3	-2	-2	-3
World	low beta	+16	+1	-24	+8	-25
	12-1 momentum	+11	-4	+7	+1	+7
World	overlay	+7	-2	-2	+2	-2
		2015-16 sell-off	Q4 2018	COVID crash	COVID rebound	2022 rate shock
Market	Book	2015-16 sell-off	Q4 2018	COVID crash	COVID rebound	2022 rate shock
US	market	-7%	-13%	-19%	+51%	-24%
	low beta	+27%	+4%	-29%	-12%	-24%
	12-1 momentum	+9%	-4%	+9%	+2%	+10%
	overlay	+8%	+2%	+2%	-4%	+4%
	market + overlay	+1%	-11%	-17%	+45%	-21%
EU	market	-13%	-12%	-21%	+38%	-26%
	low beta	+8%	+10%	-16%	-6%	-28%
	12-1 momentum	+29%	-8%	+8%	-3%	-6%
	overlay	+8%	+1%	-1%	-1%	-6%

Market	Book	2015–16 sell-off	Q4 2018	COVID crash	COVID rebound	2022 rate shock
UK	market + overlay	-5%	-11%	-22%	+37%	-30%
	market	-15%	-10%	-25%	+25%	-8%
	low beta	+12%	-13%	-12%	-2%	-17%
	12-1 momentum	+5%	-5%	+2%	-3%	+1%
	overlay	+5%	-5%	-2%	+1%	-2%
Denmark	market + overlay	-11%	-14%	-26%	+28%	-10%
	market	+3%	-12%	-13%	+53%	-24%
	low beta	+0%	-16%	-5%	-21%	+12%
	12-1 momentum	+19%	-4%	+8%	+1%	-17%
	overlay	+5%	-6%	-0%	-5%	-1%
SCANDI	market + overlay	+8%	-18%	-13%	+47%	-24%
	market	-10%	-14%	-17%	+40%	-27%
	low beta	+4%	-2%	-11%	-8%	+1%
	12-1 momentum	+15%	-4%	+5%	-2%	-9%
	overlay	+8%	-3%	-2%	-2%	-3%
World	market + overlay	-2%	-16%	-18%	+37%	-29%
	market	-14%	-13%	-23%	+49%	-28%
	low beta	+16%	+1%	-24%	+8%	-25%
	12-1 momentum	+11%	-4%	+7%	+1%	+7%
	overlay	+7%	-2%	-2%	+2%	-2%
	market + overlay	-8%	-14%	-25%	+52%	-30%

Books as in FS18: beta-neutral with the turnover buffer, FS16 costs at \$50m per book, borrow and financing; overlay at 5% volatility; market = size-weighted universe (World in USD).

- **Low beta** protects in slow sell-offs (2015–16) but not in fast crashes or rate shocks: it fell with the market in the COVID crash and in 2022 in the US, EU and World.
- **Momentum** was positive in the COVID crash in all six markets and mostly lost only a few per cent in the rebound.
- **The overlay** kept every episode within $\pm 8\%$. It adds little protection to a market portfolio; its job is return per unit of risk (FS18).

Momentum crashes (Daniel & Moskowitz 2016)

Sample	Bear months	Bear $\times$ market coefficient (t)	Market coefficient
US	3	-1.24 (t -12.3) *	-0.15
EU	5	+1.51 (t +2.4) *	-0.35
UK	20	-0.09 (t -0.3)	-0.21
Denmark	21	-0.13 (t -0.7)	-0.12

Sample	Bear months	Bear × market coefficient (t)	Market coefficient
SCANDI	8	-0.11 (t -1.3) *	-0.12
World	6	+0.16 (t +1.2) *	-0.23
America 1999–2013	51	-0.47 (t -1.6)	+0.12
Europe 1999–2013	68	-0.42 (t -1.6)	-0.10
UK 1999–2013	47	+0.17 (t +0.4)	-0.17
Denmark 1999–2013	43	-0.23 (t -1.3)	-0.01
ART pooled	209	-0.28 (t -1.9)	-0.02

*Momentum (beta-neutral, net) regressed on the equal-weight market, a bear dummy (market down over the previous 24 months) and their product; Newey–West t (ART pooled: White). A negative bear × market coefficient means momentum loses when markets rebound after a bear market. * fewer than 12 bear months: not interpretable (the US t of −12 rests on 3 months).*

2013–2026 has almost no bear-market states, so the crash risk cannot be tested there. In 1999–2013 the sign is the one Daniel & Moskowitz report (pooled coefficient -0.28, t -1.9), and FS15 showed the spring-2009 momentum loss of −42%. The risk is real, it just did not show up in our main sample.

C. Sector neutrality

Market	Sector coverage	Shortlist, unrestricted	Shortlist, within sectors (t)	Sector component	Correlation within vs unrestricted	Low beta, within sectors	Momentum, within sectors
US	95%	+4.8%	+3.4% (t +1.4)	+1.2%	0.88	+2.8%	+4.0%
EU	85%	+12.8%	+8.2% (t +3.2)	+2.6%	0.80	+10.3%	+6.0%
UK	34%	+19.5%	+15.5% (t +3.7)	+1.1%	0.89	+18.4%	+12.5%
Denmark	82%	+2.9%	-1.8% (t -0.8)	+0.2%	0.70	+1.0%	-4.5%
SCANDI	84%	+0.1%	+1.1% (t +0.5)	+2.2%	0.69	+1.2%	+0.9%
Pooled, 5 markets (primary test)	—	+8.0%	+5.3% (t +2.96)	—	—	—	—
World	76%	+8.7%	+8.2% (t +3.3)	-1.0%	0.93	+9.1%	+7.3%

Net of 10 bp per side, borrow and financing; beta-neutral legs, no buffer. Coverage: share of stocks with a sector label; the rest form one "unclassified" group. UK coverage is low because many FTSE 250 members are investment trusts missing from the sector caches. Sector component: book sorted on the sector-average signal. Pooled = US, EU, UK, Denmark and

SCANDI, the pre-registered primary test; World is left out because it overlaps. The within-sector return is 66% of the unrestricted return, above the pre-registered half.

Across the five markets, about two thirds of the shortlist's return comes from picking stocks within sectors, and the sector bets add the rest. The within-sector book is highly correlated with the unrestricted one. The exception is Denmark, where 19 stocks spread over 11 sectors leave nothing to rank within a sector. This is in line with Asness, Frazzini & Pedersen (2014), who find that low-risk returns survive industry neutralisation, and it answers the concern of Moskowitz & Grinblatt (1999) that momentum is largely an industry effect: here most of it is not.

Caveats

- Sector labels are today's, applied to all years.
- UK sector coverage is only 34%, so two thirds of UK stocks sit in one "unclassified" group; the UK within-sector result is the least sector-neutral of the five and should be read with that in mind.
- The truncation test covers the price library, not the US fundamental composites (their point-in-time logic is the filing-date lag in FS13b).
- The noise test checks the engine's timing and statistics, not the economic content of any signal.
- FS19 adds 44 trials to the ledger (now 682). The more trials a programme runs, the higher the bar for any single t-statistic (Bailey et al. 2014).

References

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planning/PREREG_FS19.md → code/fs19_engine.py (→ results/fs19_engine.json), code/fs19.py (→ results/fs19.json; uses sectors.py, fs18.py) → code/build_fs19.py. Sector source files are private.